MACKY AR · PLUGIN MANUAL · VERSION 0.1.0

StickBend User Guide

Play your keyboard. Bend with your thumb. Everything you need to install StickBend, set it up in your DAW and turn a game controller into an expressive instrument.

Install · Quick start · The interface · Pitch bend · MIDI CC · Deadzone · Stick outputs · Recording and editing · REAPER · Bitwig · FL Studio · Ableton Live · Cubase · Studio One · Troubleshooting · About the creator

Plugin: **StickBend 0.1.0** for Windows · VST3 and CLAP
Made by **Macky Ar** (Mahesh Arora) · mackyar.com
Free and open source under AGPL-3.0 · Collects no data

How to use this guide

1. Read the yellow line

Every section opens with the idea in one sentence. If you read nothing else, read that.

2. Follow the steps

Numbered steps are in the order you do them. Settings are in **bold boxes** like this.

3. Check the red box

"Watch out" boxes flag the mistakes that stop StickBend from working.

Section 1	**What StickBend does**: the idea, and where it sits in your signal chain
Section 2	**Install**: requirements, controllers that work, copying the files
Section 3	**Quick start in REAPER**: from zero to bending in five steps
Section 4	**The interface**: every control on screen, explained
Section 5	**Parameters**: the full list with ranges and defaults, send modes, useful CC numbers, deadzone
Section 6	**Stick outputs**: linking the sticks to any knob in your DAW
Section 7	**Recording and editing**: keeping every bend editable after you record
Section 8	**Other DAWs**: Bitwig, FL Studio, Ableton Live, Cubase, Studio One
Section 9	**Troubleshooting**: symptoms and fixes
Last pages	**Privacy, licence** and **about the creator**

Setups marked TESTED were checked by hand with this version. Setups marked EXPECTED follow each DAW's standard routing for plugins that output MIDI but have not been tested yet. Menu names can differ between DAW versions.

1 · What StickBend does

1.1 The idea

StickBend reads your game controller's sticks and turns them into MIDI pitch bend and CC messages, mixed in with the notes you play on your keyboard.

Your keyboard notes pass straight through untouched. StickBend only adds new messages when you move a stick, so you play with one hand on the keys and bend, swell or wobble with your thumb.

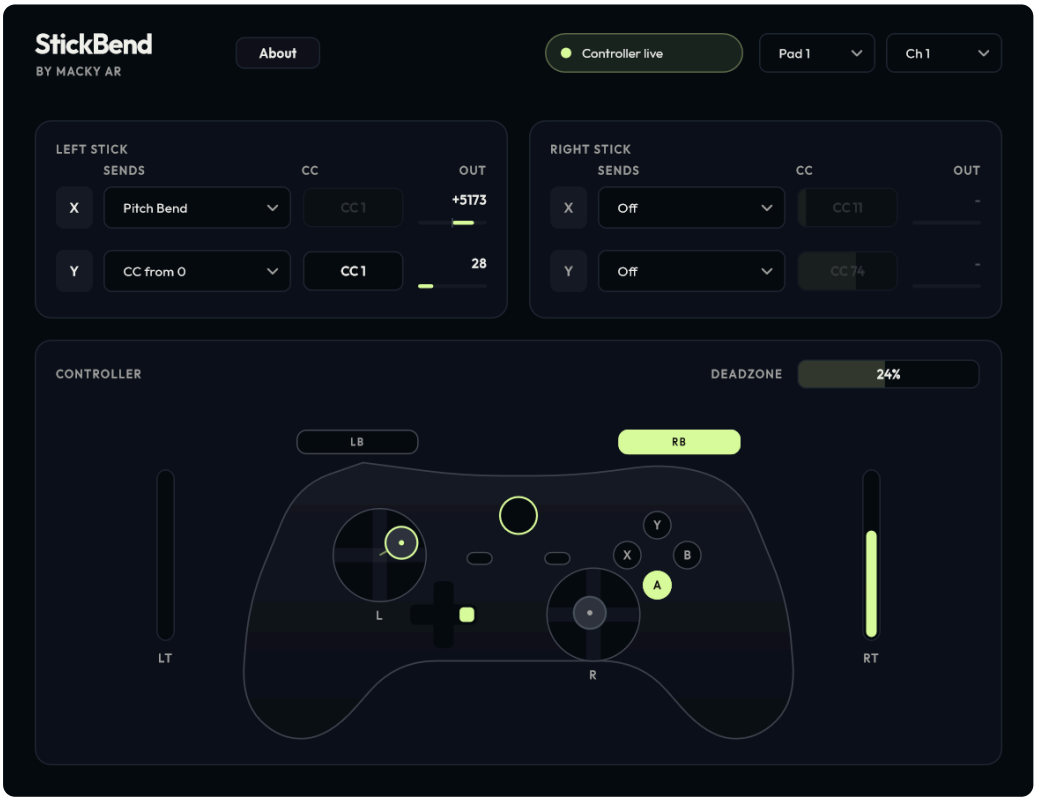
Everything it sends is ordinary MIDI. Your synth sees it exactly as if you had moved a pitch wheel or mod wheel, and your DAW records it the same way, so it stays editable afterwards.

1.2 Where it sits

StickBend always goes before your instrument, never after it.

MIDI keyboard	→	StickBend	→	Your synth	→	Audio out
notes		notes + bend + CC		plays it all		speakers

Watch out: if StickBend is placed after the synth, the synth never receives the bends and you hear nothing change.



StickBend with the left stick pushed up and right, RT half pressed, and RB, A and D-pad right held.

2 · Install

2.1 What you need

Computer	Windows 10 or 11, 64-bit
DAW	Any host that loads VST3 or CLAP plugins and accepts MIDI from plugins: REAPER, Bitwig Studio, FL Studio, Cubase, Studio One, Ableton Live and others
Controller	An Xbox-compatible (XInput) controller, wired, wireless adapter or Bluetooth
Keyboard	Any MIDI keyboard. StickBend also works with no keyboard at all, for drawing in bends over existing MIDI.

2.2 Controllers that work

If Windows sees it as an Xbox controller, StickBend can use it.

- **Works directly:** Xbox One and Xbox Series controllers, Xbox 360 controllers, most third-party PC pads, 8BitDo pads in X-input mode.
- **Works with a helper:** PlayStation DualShock 4 and DualSense, using the free tool DS4Windows, which makes them appear as Xbox controllers.
- **Not yet:** Nintendo Switch Pro and pads that only speak DirectInput.

> **Tip:** up to four controllers can be connected. Each gets a slot from 1 to 4, and the **Pad** menu in StickBend picks which one to listen to.

2.3 Copy the files

1. Close your DAW.
2. Copy **StickBend.vst3** into **C:\Program Files\Common Files\VST3**
3. Copy **StickBend.clap** into **C:\Program Files\Common Files\CLAP**
4. Open your DAW and rescan plugins. StickBend appears under the vendor **Macky Ar**.

> **Watch out:** Windows locks plugin files while a DAW has them open. Always close the DAW before replacing StickBend with a newer version, or the copy will fail.

Which format? Use CLAP if your DAW supports it (REAPER, Bitwig). It carries MIDI natively. Use VST3 everywhere else.

3 · Quick start in REAPER TESTED

3.1 Five steps to your first bend

Put StickBend in the track's Input FX, your synth in the normal FX chain, arm the track and move the stick.

1. Create a track and add your synth to its normal FX chain as usual.
2. Click the small `IN FX` button on the track (Input FX) and add **StickBend**.
3. Set the track's record input to your MIDI keyboard, then **arm** the track and turn **record monitoring** on.
4. Check the status pill at the top of StickBend says **Controller live**.
5. Hold a note and push the left stick sideways. The pitch bends. Push it up or down and the mod wheel moves.

3.2 Why Input FX?

REAPER records what comes **out of the Input FX**. With StickBend there, every stick move is recorded into your MIDI item as real pitch bend and CC data that you can edit later.

If you put StickBend in the normal FX chain instead, you still hear the bends live, but REAPER records only your keyboard, so the bends vanish from the recording.

> **Watch out:** Input FX only run while the track is armed with monitoring on. If StickBend seems dead, check the arm button first.

3.3 Default mapping

AXIS	SENDS	EFFECT
Left stick X	Pitch Bend	Left bends down, right bends up
Left stick Y	CC 1, from 0	Mod wheel. Up or down raises it, centre returns to 0
Right stick X	Off (CC 11 ready)	Turn on for expression
Right stick Y	Off (CC 74 ready)	Turn on for brightness or filter cutoff

4 · The interface

4.1 Header

Status pill	**Controller live** (lime dot) when the selected pad is connected. **No controller** when it is not. StickBend checks again about once a second, so you can plug in a pad at any time.
Pad	Which controller slot (1 to 4) to read. Slot 1 is the first pad Windows connected.
Ch	The MIDI channel (1 to 16) for the bend and CC messages. Match it to the channel your synth listens on. Most synths listen on all channels, so 1 is usually fine.
About	Version, credits, links and the privacy summary.

4.2 Stick cards

One card per stick, one row per axis. Each row has three parts:

Sends	What this axis sends: **Off**, **Pitch Bend**, **CC centred** or **CC from 0**. Explained in section 5.2.
CC	The CC number, 0 to 127. Only active when Sends is one of the CC modes, otherwise it is dimmed.
Out	The value being sent right now, with a small meter. Pitch bend shows -8192 to +8191 (0 is centre). CC shows 0 to 127.

4.3 Controller visualizer

The picture mirrors your real controller, so you can see exactly what StickBend is reading.

- **Sticks** move live. The knob turns lime when it leaves the deadzone and a line shows the direction.
- **The shaded cross** inside each stick is the deadzone. Inside the vertical band, X sends nothing. Inside the horizontal band, Y sends nothing.
- **Triggers, bumpers, face buttons, D-pad, Back, Start** and **stick clicks** light up when pressed, handy for checking the pad works.
- **Deadzone** slider at the top right of this panel. Drag it or double-click to type a value.

5 · Parameters

5.1 Full parameter list

These are the names you see in your DAW's automation and parameter lists.

PARAMETER	RANGE	DEFAULT	WHAT IT DOES
Controller	1 to 4	1	Controller slot to read
MIDI Channel	1 to 16	1	Channel for outgoing bend and CC
Deadzone	0 to 50%	24%	How far a stick must move before it sends anything
Left X / Left Y / Right X / Right Y sends	Off, Pitch Bend, CC centred, CC from 0	Bend, CC from 0, Off, Off	What each axis sends
Left X / Left Y / Right X / Right Y CC #	0 to 127	1, 1, 11, 74	CC number when the axis is in a CC mode
Stick Left X / Left Y / Right X / Right Y	0 to 1	0.5	Follow the sticks while you play. When the stick rests, automation on these plays back as bends and CCs (sections 6 and 7).

5.2 The four send modes

Off	The axis sends nothing.
Pitch Bend	Full 14-bit pitch bend, the same message a pitch wheel sends. Centre is no bend. **How far** the note bends (2 semitones, 12, and so on) is set in your synth, not in StickBend.
CC centred	Rests at 64. One direction goes down to 0, the other up to 127. Good for things that have a middle: pan (CC 10), or sweeping a filter both ways.
CC from 0	Rests at 0. Pushing either way raises it to 127, so the stick works like a spring-loaded mod wheel. Good for vibrato depth (CC 1).

Watch out: pitch bend not bending far enough, or at all? Check the synth's own pitch bend range. If it is set to 0, StickBend's bends do nothing.

5.3 Useful CC numbers

CC 1 · Mod wheel	Vibrato or the synth's main modulation. The classic choice.
CC 2 · Breath	Expression in many wind and string libraries
CC 7 · Volume	Channel volume
CC 10 · Pan	Use with CC centred
CC 11 · Expression	Swells and dynamics in orchestral libraries
CC 71 · Resonance	Filter resonance on many synths
CC 74 · Brightness	Filter cutoff on many synths

Watch out: CC 7 or CC 11 in **CC from 0** mode drops the volume to zero whenever the stick is centred. Use them with care, or pick CC centred.

5.4 Deadzone

The deadzone stops a resting stick from sending tiny wobbles, so your notes stay perfectly in tune when you let go.

Real sticks never rest exactly at the centre. The deadzone ignores small movements near the middle, then rescales the rest so a full push still reaches the full value. Each axis has its own deadzone, which is why it is drawn as a cross.

- **15 to 30%** suits most controllers. The default 24% matches Microsoft's recommendation.
- Raise it if notes drift slightly sharp or flat when the stick is untouched.
- Lower it for finer, earlier response on a new controller.

6 · Stick outputs

6.1 Control any knob

The four Stick parameters follow the sticks, so you can link them to any knob on any plugin.

StickBend exposes **Stick Left X**, **Stick Left Y**, **Stick Right X** and **Stick Right Y** as plugin parameters. Each rests at 0.5 when the stick is centred and moves from 0 to 1 across its full travel. They move whatever the Sends setting is, even if the axis is Off.

They also work the other way. While a stick is at rest, StickBend **plays whatever its Stick parameter says**, so recorded or hand-drawn Stick envelopes turn into real pitch bend and CC on playback. Move the stick and it takes over again.

In REAPER: parameter modulation EXPECTED

1. Open the plugin you want to control, for example a reverb on the same track.
2. Click the knob you want, then click **Param** in the FX window and choose **Parameter modulation/MIDI link**.
3. Tick **Link from MIDI or FX parameter** and pick **StickBend: Stick Right Y**.
4. Adjust the scale and offset so the knob moves over the range you want.

> **Tip:** to drive a plugin on another track, use the Stick outputs as automation (next section) or send StickBend's MIDI to that track instead.

7 · Recording and editing

7.1 Two ways to record, both editable

Record the stick as MIDI data in the clip, or as automation envelopes. Either way you can redraw it later.

	AS MIDI DATA	AS AUTOMATION
What is recorded	Pitch bend and CC messages inside the MIDI item	Envelopes for the Stick parameters, which StickBend plays back as bends and CCs
Where you edit it	The MIDI editor's CC lanes (Pitch, CC 1 and so on)	The track's automation lanes
Best for	Bends and vibrato that belong to the performance	Reshaping a performance as smooth curves, and driving other plugins' knobs
How in REAPER TESTED	StickBend in Input FX, arm, record	StickBend in the normal FX chain, automation mode **Latch** or **Write**, arm the Stick envelopes, play. Switch back to **Read** to hear it. To record with the StickBend window closed, see section 9.

7.2 Playing back without fighting your edits

StickBend only sends a message when a stick **moves**. A centred stick is completely silent, so on playback your recorded and edited bends play exactly as written.

> **Watch out:** if you touch the stick during playback with StickBend still active on the track, the new movement mixes with the recorded data. Disarm the track (Input FX) or bypass StickBend while you listen back.

7.3 Editing tips

- Select a stretch of CC points in the lane and delete or redraw it with the pencil to fix a single bend without replaying the take.
- Draw a straight line back to centre at the end of a bend to make sure the next note starts in tune.
- Copy a vibrato shape you like and paste it under other notes.

8 · Other DAWs

The idea is the same everywhere: StickBend's MIDI output must reach your synth. These are the standard routings for each DAW. They are marked EXPECTED because they have not been tested with this version yet.

8.1 Bitwig Studio EXPECTED

Load the **CLAP** version. On an instrument track, drop StickBend into the device chain **before** your instrument. Notes and bends flow straight into it. Record as normal and edit bends in the clip's expression lanes.

8.2 FL Studio EXPECTED

1. Load StickBend (VST3) in the Channel Rack. Open its wrapper settings (gear icon) and set an **Output port**, for example 1.
2. Load your synth and set its **Input port** to the same number.
3. Select StickBend's channel and play. Your keyboard goes in, the synth plays with bends.

8.3 Ableton Live EXPECTED

1. Put StickBend (VST3) on a MIDI track that receives your keyboard. Set monitoring to **In**.
2. On a second MIDI track with your synth, set **MIDI From** to the first track and choose **StickBend** in the second menu.
3. Arm the synth track to record the bends with your notes.

8.4 Cubase and Studio One EXPECTED

Load StickBend (VST3) as an instrument on its own track that receives your keyboard. Then set the input of your synth track to StickBend's MIDI output. Look for "plugin MIDI output" in your DAW's manual if the menus differ.

> **Tip:** tried StickBend in one of these DAWs? Send the steps that worked to macky.business725@gmail.com and they will go into the next version of this guide.

9 · Troubleshooting

SYMPTOM	FIX
Status says No controller	Check the pad is on and connected. Try the other slots in the **Pad** menu. PlayStation pads need DS4Windows running.
Visualizer moves but nothing bends	StickBend must be before the synth. In REAPER use Input FX with the track armed and monitoring on. Check the synth's pitch bend range is not 0.
Live bends work but disappear from the recording	In REAPER, StickBend is in the normal FX chain, so only your keys are recorded into the item. Move it to Input FX, or record the Stick envelopes instead (section 7.1).
Envelope does not record	Set the track to **Latch** or **Write** and arm the Stick envelope. Use version 0.1.0 or later.
Envelope only records while the StickBend window is open	REAPER ignores parameter changes from plugins with closed windows by default. In **Preferences** → **Plug-ins** → **VST**, under VST compatibility, set **Parameter automation notifications** to **process all notifications**. Or keep the window open, or record as MIDI through Input FX instead.
Envelope recorded but playback has no bend	Check the axis's **Sends** is still Pitch Bend or CC, the envelope isn't bypassed, and the track is in **Read** or **Latch** mode rather than Trim/Read. Leave the stick untouched during playback.
Notes slightly out of tune at rest	Raise the **Deadzone**. Your stick rests a little off centre.
Synth ignores the bends	Match the **Ch** setting to the channel your synth listens on.
Volume drops to silence	An axis is sending CC 7 or CC 11 in CC from 0 mode. Change the CC number or use CC centred.
Cannot replace the plugin file	Close the DAW first. Windows locks plugins that are in use.
VST3 version misbehaves in your DAW	Try the CLAP version if your DAW supports it.

Still stuck? Email macky.business725@gmail.com with your DAW name and version, your controller model, and what the status pill says.

Privacy and licence

P Privacy

StickBend collects nothing and never goes online by itself.

No accounts, no analytics, no telemetry, no licence checks. Controller readings are turned into MIDI on the spot and never stored or sent anywhere. Your settings are saved by your DAW inside your project. The only time anything goes online is when you click a link in the About panel, which opens your own browser. The full policy is in **PRIVACY.md**.

L Licence

StickBend is free and open source under the **GNU Affero General Public License v3**. You can use it in any project, commercial or not, at no cost. Copyright (C) 2026 Mahesh Arora (Macky Ar). It comes with no warranty.

Built with JUCE (AGPL-3.0), the VST3 SDK, CLAP and clap-juce-extensions (all MIT) and the Outfit typeface (SIL Open Font License). Full notices are in THIRD_PARTY_NOTICES.md. VST is a registered trademark of Steinberg Media Technologies GmbH.

ABOUT THE CREATOR

Macky Ar

StickBend was built by musician **Macky Ar** (Mahesh Arora), a multi-instrumentalist who wanted pitch bend to feel as natural as moving a thumb.

Pitch wheels are small, stiff and stuck at the far left of the keyboard. A game controller is cheap, familiar and comfortable to hold, and almost everyone already knows how to use one. StickBend turns it into an accessible expression tool, so bends, swells and vibrato are within reach of anyone, on any keyboard, including players who find a pitch wheel awkward to reach or grip.

WEBSITE
mackyar.com

SPOTIFY
Macky Ar on Spotify

BANDCAMP
mackyar.bandcamp.com

YOUTUBE
@MackyAr

INSTAGRAM
@macky_ar

EMAIL
macky.business725@gmail.com